Baseball Product — API testing and product validation report

Scope: small-scale functional and manual validation of the Baseball Product against the deployed SportsAPI365 baseball API. This is **not** production certification, load testing, or a security assessment.

Statuses used: `PASS` / `FAIL` / `BLOCKED` / `NOT TESTED` / `NEEDS INVESTIGATION`. A case is marked executed only when repo evidence or a read-only GET from this review exists. Documented claims that were not independently observed are labeled as documented.

A. Document overview

Field	Value
Product	Baseball Product (`/Users/apple/Desktop/Baseball_Product`)
App	React UI `apps/web` (Vite, port **5175**). No local baseball backend.
API	`GET https://sports-api.sportsdata365.com/msdata/v1/baseball/*`
Catalog	**42** routes in `apps/web/src/catalog.ts` and `data/baseball-routes.json` (`count` : 42)
Client	`apps/web/src/msdata.ts` — browser calls same-origin `/msdata` ; Vite proxy and Vercel handlers attach keys server-side
Config names (values not recorded)	`MSDATA_BASE_URL` , `MSDATA_API_KEY` , `MSDATA_LIVE_KEY` (optional `WEB_PORT`)
This report	`docs/api-testing-report.md` (new). Does **not** replace existing reports listed in `docs/README.md` .
Review date	21 Oct 2026
Nature	Evidence synthesis + a small set of **additional read-only GETs** (section O). No POST/PUT/DELETE. No load/stress. No app functionality changes.

Evidence generations used

Artifact	Generated	Role
`data/baseball-endpoint-evidence.json`	2026-10-21T12:31:40.219870+00:00	Live probe of all 42 catalog routes + 11 extra probes
`docs/baseball_api_testing_report.md`	same timestamp	Generated matrix from that JSON
`data/baseball-variation-tests.json`	2026-10-21T12:33:50.939014+00:00	51 variation/pagination/status tests
`docs/baseball_api_variation_testing.md`	same timestamp	Variation write-up
`data/full-catalog-test.json`	lab sweep; `rowCount` 71, `catalogCovered` 42	Broader catalog + error/paging rows
`data/calls.json`	196 recorded calls, timestamps 12:57–13:01Z	In-product / lab HTTP log
`data/findings.json` / `data/gaps.json`	same lab window	Structured findings

Artifact	Generated	Role
`docs/Baseball_Product_Review.md`, `docs/baseball_api_improvements.md`	Checked 21 Oct 2026	Playbook + API-team list; UI walk notes
Additional read-only GETs (this review)	21 Oct 2026, after the files above	Re-check of listed observations (section O)

Sample IDs (verified in evidence `discoveredIds` and this review's GETs): event `67599823`, Yankees `84917477`, Ohtani `71698966`, MLB `53168725`, season `68144200`, manager `62643844`. Probe also recorded lineup player `39625815`, enrichment type `h2h_events`, team enrichment `events_last`. Variation tests added canceled `82011329` and not-started `28606880`.

B. Executive summary

The catalog of **42** baseball routes was executed. **40** returned HTTP 200 on the sample entities; **2** catalog routes returned **404** on valid Yankees / finished-game IDs (`best-players`, team `rankings`). No 5xx appeared in `calls.json` (196 calls: 181×200, 8×400, 7×404).

The product can ship **directory, search, match-centre extras, team profile, player profile, and MLB total standings**. It cannot ship a trustworthy **Live** or **Results** feed from `GET /events` alone: `status=finished` and `status=inprogress` both returned **`total=0`** while `GET /team/84917477/events/last` returned **`total=2123`** in the same window (re-confirmed this review). Matches → Results therefore falls back to Yankees `events/last` (code + `calls.json`).

List rows on `/events` and `/fixtures/today` use a **string** `status` and **`{ series }`** scores. Event **detail** for `67599823` uses a **status object** `{ code: 100, type: "finished", description: "Ended" }` and **`inningsBaseball` { run, hits, errors }** (Yankees 3 / 7 / 0 vs Rays 4 / 7 / 1). In this review, `/fixtures/today` and `/events` (`limit=20`) returned the **same 18 IDs** and the same 5290-byte bodies.

Other confirmed gaps: `/h2h` is 200 with `teamDuel` / `managerDuel` null; standings `home` / `away` are 200 with empty `rows` (total has **90**); `page > totalPages` is 200 with an empty list while `total` stays high; `X-Live-Key` alone is 200 and `x-apim-key` alone is 403; football-style **player season statistics** returns 200 for Ohtani but is **not in the 42-route catalog** and is not called by `Player.tsx`.

This report does **not** assign numeric experience scores. Existing `docs/baseball_api_reports.md` contains lab scores; they are out of scope here.

C. Product overview + workflow-to-endpoint table

What the product is (from repo, not assumed): a React lab at `http://127.0.0.1:5175` that calls deployed `/v1/baseball/*` through a same-origin `/msdata` proxy (`apps/web/vite.config.ts` locally; `apps/web/api/msdata/[...path].ts` and `apps/web/middleware.ts` on Vercel). Keys stay in server env. Footer copy: "SportsAPI365 baseball · no local API".

Routes in the chrome (`apps/web/src/components/Layout.tsx`): Home `/`, Matches `/matches`, Competitions `/competitions`, Teams `/teams`, Players `/players`, Search `/search`, "42 routes" `/endpoints`. Additional routes exist but are not in the header: `/match/:id`, `/player/:id`, `/team/:id`, `/manager/:id`, `/competition/:id` (`App.tsx`).

Documented vs observed: screens and tabs below are from **source**. What the user saw on 21 Oct 2026 is taken from `data/calls.json`, `data/findings.json`, and `docs/baseball_api_improvements.md` where those files name the screen. No screenshots were created for this report.

Feature / screen	Endpoints (from source)	Expected (product intent in source/docs)	Observed	Finding
Home `/`	`fixtures/today`, `events`, `tournaments` (`Home.tsx`)	Sections Live / Today / Upcoming / Completed + competition cards	**Code:** Live = events filtered to status label Live; Today = fixtures list; Upcoming/Completed = subsets of today. **API:** `/events` and `/fixtures/today` are the same 18 `notstarted` rows (this review). `status=inprogress` total=0 → Live empty. List scores are `series` only. **Lab 12:58Z:** `calls.json` Home "Live match card" also issued N+1 `GET /event/{id}` with `fallbackUsed: true` (8 extraCalls in findings). **Current `Home.tsx` does not N+1.**	Live empty; Today cards cannot show R/H/E from the list payload. Distinguish current source vs earlier lab N+1.
Matches → Live	`GET /events?status=inprogress`	Live games	200, `total=0` (evidence extra probe + variation + this review)	Results/Live blocker. `FAIL`
Matches → Today	`GET /fixtures/today`	Today's schedule	200, `total=18`, same IDs as `/events` (this review)	Identical to default `/events` in this run. `NEEDS INVESTIGATION` (semantics)
Matches → Upcoming	`GET /events?status=notstarted`	Upcoming	200, `total=18` (variation + this review)	Works for the current thin feed. `PASS`
Matches → Results	`GET /events?status=finished`; if empty, `GET /team/84917477/events/last` (`Matches.tsx` + `SAMPLE.teamId`)	Completed games	finished `total=0`; fallback 200, Yankees `total=2123` (`calls.json` 12:58:42–43Z; note in UI)	Fallback is Yankees-only and mixes canceled (`82011329` first row this review, status object type `canceled`). `FAIL`
Match centre `/match/:id`	`GET /event/{id}` plus tabs: at-bats, lineups, statistics, player-statistics, top-performers, umpires, h2h, best-players, pregame-form, team-streaks, weather, enrichments (`Match.tsx`)	Full game	Sample `67599823`: detail 200 with R/H/E; best-players **404**; h2h **200 null duels**; other extras 200. Lab recorded Match Centre calls for header / streaks / weather / pregame / at-bats / top-performers / umpires — not every tab.	Centre usable except h2h empty and best-players 404.

Feature / screen	Endpoints (from source)	Expected (product intent in source/docs)	Observed	Finding
Search `/search`	`GET /search?q=` (`Search.tsx`)	Typed hits → player/team/competition	`q=Yankees` → 1 team hit id `84917477`. Catalog audit: Ohtani 200 (249 B), MLB 200 (444 B). Missing `q` → 400, `message` is an **array**. Playbook typo `Yankeess`: **no JSON evidence**.	Required `q` works. Typo case `NOT TESTED`.
Players directory	`GET /players?page&limit=20`	Paginated directory	Probe: total **9758**, totalPages **1952**. UI pages on `hasMore`.	`PASS`
Player `/player/:id`	`player/{id}`, `events/last`, `events/next`	Profile + recent/upcoming	Ohtani 200; `team: null`; last total **1253**; next **total=0**. Season stats route not called. UI lab: overview only.	Profile works; no season line; upcoming empty.
Teams directory	`GET /teams`	Directory	Probe total **255**, totalPages **51**.	`PASS`
Team `/team/:id`	profile, players, events last/next, featured-event, near-events, performance, rankings, unique-tournaments, standings-seasons, media, enrichments	Team hub	Yankees 200; roster `players` n=48; rankings **404**; events/next total=0; featured-event is finished `67599823` (findings). UI lab hit overview/featured/near/performance only.	Rankings 404 looks like a missing team.
Manager `/manager/:id`	`GET /manager/{id}`	Profile	200 `{ id: 62643844, name: "Kevin Cash", imageUrl }` (image host path only). Not in header; “open from a lineup” (playbook). No `calls.json` row with `screen: Manager`.	API `PASS`; UI walk for this screen `NOT TESTED` in `calls.json`.
Competitions `/competitions`	`GET /tournaments`	Directory	Probe total **84**.	`PASS`
Competition `/competition/:id`	unique-tournament, seasons, events, scheduled-events/{date}, standings/{type} total|home|away, round/{round}	MLB hub	MLB 200 name `MLB`; seasons total 24; events total **29267**; scheduled 2026-10-21 total **1**; standings total **90** rows; home/away **0** rows. UI lab: overview, seasons, events.	Home/away tables empty.
42 routes `/endpoints`	Every `CATALOG` path (`EndpointLab.tsx`) with sample IDs	Click-to-GET lab	Source fires GETs; sample IDs match probe. No dedicated `calls.json` screen name for this page; catalog sweep covers the same paths.	Lab page exists; sweep `PASS` as API coverage.

D. API inventory (42 catalog routes + extra observed)

Base path: `https://sports-api.sportsdata365.com/msdata` + `/v1/baseball/...`. Auth headers used by probes and the Vite proxy: `x-apim-key` (from `MSDATA_API_KEY`) and `X-Live-Key` (from `MSDATA_LIVE_KEY`).

IDs in the 12:31Z probe: event `67599823`, team `84917477`, player `71698966` (event player-stats used lineup `39625815`), manager `62643844`, tournament `53168725`, season `68144200`, date `2026-10-21`, enrichments `h2h_events` / `events_last`.

D1. Catalog (42)

#	Route	Called path (probe)	HTTP	Kind / list key	Count / pagination	Inventory note
1	`/event/{eventId}`	`event/67599823`	200	match `event`	—	Detail. Re-GET: status object; `inningsBaseball`.
2	`/event/{eventId}/at-bats`	`.../at-bats?page=1&limit=5`	200	`atBats`	n=5, total=70, pages=14	Empty 200 on canceled/not-started (variation).
3	`/event/{eventId}/best-players`	`.../best-players`	**404**	error envelope	—	`ENRICHMENT_NOT_FOUND` on finished sample.
4	`/event/{eventId}/enrichments`	`.../enrichments`	200	`types`	n=5	h2h_events, highlights, series, team_streaks, votes.
5	`/event/{eventId}/enrichments/{type}`	`.../enrichments/h2h_events`	200	`events`	n=10, total=219	
6	`/event/{eventId}/h2h`	`.../h2h`	200	`teamDuel`, `managerDuel`	both **null**, 36 B	
7	`/event/{eventId}/lineups`	`.../lineups`	200	`home`, `away`	home `dressedPlayers` 16; `manager` null	
8	`/event/{eventId}/player-statistics`	`.../player-statistics?page=1&limit=5`	200	`playerStatistics`	n=5, total=32, pages=7	
9	`/event/{eventId}/player/{playerId}/statistics`	`event/67599823/player/39625815/statistics`	200	player object	—	Match-level, not season.
10	`/event/{eventId}/pregame-form`	`.../pregame-form`	200	`homeTeam`, `awayTeam`	form arrays + eventIds	
11	`/event/{eventId}/statistics`	`.../statistics`	200	`statistics`	n=1 period ALL	Fielding + batting groups.
12	`/event/{eventId}/team-streaks`	`.../team-streaks`	200	six keys, all `[]`	111 B	Probe `PASS_PARTIAL`. Not-started sample 404 in lab.
13	`/event/{eventId}/top-performers`	`.../top-performers`	200	`topPerformers`	n=3, roles batter	

#	Route	Called path (probe)	HTTP	Kind / list key	Count / pagination	Inventory note
14	`/event/{eventId}/umpires`	`.../umpires`	200	`umpires`	n=6	HP/1B/2B/3B/LF/RF.
15	`/event/{eventId}/weather`	`.../weather`	200	`conditions`	n=2	
16	`/events`	`events?page=1&limit=5`	200	`items`	n=5, **total=18**, pages=4	List shape ≠ detail.
17	`/fixtures/today`	`fixtures/today?page=1&limit=5`	200	`items`	n=5, **total=18**, pages=4	Same 18 IDs as `/events` this review (`limit=20`).
18	`/manager/{managerId}`	`manager/6264384`	200	`manager`	id, name, imageUrl	
19	`/player/{playerId}`	`player/7169896`	200	`player`	`team: null`	
20	`/player/{playerId}/events/{direction}`	`player/7169896/events/last`	200	`events`	n=5, total=1253, pages=251	`next` extra: total=0.
21	`/players`	`players?page=1&limit=5`	200	`items`	n=5, total=9758, pages=1952	
22	`/search`	`search?q=Yankees`	200	`items`	n=1, total=1	type/id/name.
23	`/team/{teamId}`	`team/84917477`	200	`team`	`country: null` ; logoUrl present	
24	`/team/{teamId}/enrichments`	`.../enrichments`	200	`types`	n=8	
25	`/team/{teamId}/enrichments/{type}`	`.../enrichments/events_last`	200	`events`	n=10, total=30, pages=3	
26	`/team/{teamId}/events/{direction}`	`team/84917477/events/last`	200	`events`	n=5, **total=2123**, pages=425	`next` extra: total=0.
27	`/team/{teamId}/featured-event`	`.../featured-event`	200	match `event`	finished `67599823` (findings)	Not “currently live”.
28	`/team/{teamId}/media`	`.../media?page=1&limit=5`	200	`media`	n=5, total=10, pages=2	
29	`/team/{teamId}/near-events`	`.../near-events`	200	`previousEvent` , `nextEvent`	—	
30	`/team/{teamId}/performance`	`.../performance?page=1&limit=5`	200	`events` (+ form, points)	n=5, total=20, pages=4	
31	`/team/{teamId}/players`	`.../players`	200	`players`	n=48	Nested `player` objects.

#	Route	Called path (probe)	HTTP	Kind / list key	Count / pagination	Inventory note
32	`/team/{teamId}/rankings`	`.../rankings`	**404**	error envelope	—	`ENRICHMENT_NOT_FOUND`.
33	`/team/{teamId}/standings-seasons`	`.../standings-seasons`	200	`uniqueTournamentSeasons`	n=2, total=4, pages=1	
34	`/team/{teamId}/unique-tournaments`	`.../unique-tournaments`	200	`uniqueTournaments`	n=3, total=3	
35	`/teams`	`teams?page=1&limit=5`	200	`items`	n=5, total=255, pages=51	
36	`/tournaments`	`tournaments?page=1&limit=5`	200	`items`	n=5, total=84, pages=17	
37	`/unique-tournament/{tournamentId}`	`unique-tournament/53168725`	200	`uniqueTournament`	name MLB	full-catalog: `currentSeasonId` missing on object; seasons envelope has it.
38	`/unique-tournament/{tournamentId}/events`	`.../events`	200	`events`	n=10, **total=29267**, pages=2927	Page 2000 is real data (variation).
39	`/unique-tournament/{tournamentId}/scheduled-events/{date}`	`.../scheduled-events/2026-10-21`	200	`events`	n=1, total=1	
40	`/unique-tournament/{tournamentId}/season/{seasonId}/round/{round}`	`.../round/1`	200	`events`	n=10, total=2458, pages=246	
41	`/unique-tournament/{tournamentId}/season/{seasonId}/standings/{type}`	`.../standings/total`	200	`standings`	1 table, **90** rows	home/away extra: 0 rows.
42	`/unique-tournament/{tournamentId}/seasons`	`.../seasons?page=1&limit=5`	200	`seasons`	n=5, total=24, pages=5	`currentSeasonId` on envelope.

D2. Extra observed routes / calls labeled **not-in-catalog**

These are **not** in `CATALOG` / `baseball-routes.json`. They were still observed.

Path	HTTP	Source	Label
`/v1/baseball/player/71698966/unique-tournament/53168725/season/68144200/statistics`	**200**	Additional GET this review (also documented 21 Oct in `baseball_api_improvements.md`)	**not-in-catalog**. Keys include `player`, `team`, `statistics`, `playerId`, `teamId`, `uniqueTournamentId`, `seasonId`. Nested: `battingAvg` 0.275, `battingHomeRuns` 30, `battingAtBats` 523, `appearances` 135, `battingHits` 144.
`/player/{id}/events/sideways`	400	`full-catalog-test.json` / `calls.json`	Invalid `direction` on a catalog template, not a new route.
`/unique-tournament/.../standings/form`	400	same	Invalid standings `type`.

Path	HTTP	Source	Label
`/unique-tournament/.../scheduled-events/not-a-date`	400	same	Invalid date.
`/event/abc`, `/event/999999999`, `/player/999999999`, `/team/999999999`	400 / 404	catalog extras	Invalid/unknown IDs on catalog templates.

`docs/baseball_api_endpoint_catalog.md` notes basketball-only `incidents`, `injuries`, and season-statistics paths as **absent from the baseball catalog**. Season statistics is nonetheless **reachable** (row above). Incidents/injuries were **not** called in this review (`NOT TESTED`).

E. Response schema and data quality

Documented (catalog/improvements) vs observed (JSON + this review's GETs). Truncated `bodyPreview` strings in evidence are not treated as full payloads; this review's event-detail GET is used for score/status fields that the 1200-character preview cut off.

E1. Event list vs event detail (confirmed)

`GET /events` and `GET /fixtures/today` row 0 (id `28606880`, this review):

```
{
  "id": 28606880,
  "status": "notstarted",
  "homeScore": { "series": 4 },
  "awayScore": { "series": 2 },
  "tournamentId": 53168725,
  "competitionName": "MLB Postseason",
  "seasonId": 68144200
}
```

Row keys: `id, tournamentId, homeTeam, awayTeam, startTimestamp, status, homeScore, awayScore, seasonId, competitionName`. No `tournament` object. `status` is a **string**. `startTimestamp` is unix seconds (probe row and findings).

`GET /event/67599823` (this review, 4959 B, same size as 12:31Z probe):

- Teams: New York Yankees vs Tampa Bay Rays; venue `Yankee Stadium`.
- `status`: `{ "code": 100, "type": "finished", "description": "Ended" }`; `statusType`: `"finished"`.
- `homeScore` keys include `series, current, display, innings, period1–period9, normaltime, inningsBaseball`.
- `homeScore.inningsBaseball`: `{ "run": 3, "hits": 7, "errors": 0 }`; away `{ "run": 4, "hits": 7, "errors": 1 }`.
- `tournament.id`: **null**; `tournament.name`: `"MLB Postseason"`; nested `uniqueTournament.id`: **53168725**, name `"MLB"`.
- `startTimestamp`: integer `1791417600`.

`apps/web/src/msdata.ts` `asMatch` reads `uniqueTournament` **or** `tournament` **or** nested uniqueTournament. Because `tournament` exists, `tournament.id` (null) wins — Match → Competition link can fail even though nested MLB id is present. `findings.json` `BROKEN_RELATIONSHIP` on `uniqueTournament.id` matches this client mapping.

Team `events/last` uses list key **`events`** (not `items`) and a **status object**. First row this review: id `8201132 9`, `{ code: 70, type: "canceled", description: "Canceled" }`, `homeScore` keys `["series"]` only.

E2. List-key split (observed)

Family	Array key	Pagination fields
`/events`, `/fixtures/today`, `/teams`, `/players`, `/tournaments`, `/search`	`items`	`page, limit, total, totalPages, hasMore`
team/player/unique-tournament event lists, at-bats, media, etc.	`events` / `atBats` / `media` / …	same pagination fields when present
team roster	`players`	no page fields in probe
standings	`standings[].rows`	no page fields

E3. Other shapes (sanitized)

Resource	Observed
Search hit	`{ "type": "team", "id": 84917477, "name": "New York Yankees", "countryAlpha2": null, "gender": "M" }`
H2H	`{ "teamDuel": null, "managerDuel": null }`
Team streaks (finished sample)	all six buckets `[]`
Top performers	3× `role: "batter"` with battingHits / battingAtBats / battingHomeRuns / battingRbi
Umpires	assignment HP, 1B, 2B, 3B, LF, RF
Weather	temperature, humidity, windSpeed, isForecast true/false
Pregame form	`form: ["L","L","W","W","W"]` home; eventIds listed; `source: "derived"`
Lineups home	`confirmed: true`, `startersListed: 11`, `dressedPlayers: 16`, `manager: null`, `formation: null`; player rows have `battingOrder`, `appearanceRoles`, `startingPitcher`
Player Ohtani	position `DH`, jersey `17`, height 193, `team: null`
Team Yankees	`country: null`, `manager: null`, `logoUrl` present (path only)
Standings total row keys	team, position, matches, wins, draws, losses, scoresFor/Against, gamesBehind, winPercentage, group, …
Error envelope	`statusCode, message, error, path, timestamp, requestId` (+ `details` on 404)
Validation `message`	sometimes **string**, sometimes **array of strings** (search missing `q`; `status=ended`)

E4. Quality issues tied to evidence

- Finished Yankees–Rays has a full baseball box on **detail**, not on **list**.

- Ohtani `team` is null on the player object (findings `PREFERRED_FIELD_MISSING`).
- Yankees team `country` is null on `/team/{id}` while the same club nested on event detail has `country: { name: "USA", alpha2: "US" }`.
- Shirt numbers on the lineup preview rows shown are null.
- `calls.json` timestamps are unix-ms ISO; API `startTimestamp` is unix **seconds** (findings `SCHEMA_INCONSISTENCY`).

F. Manual API test strategy

Goal: confirm the 42 catalog routes respond, that product screens can consume them, and that known gaps still reproduce. Not a performance SLA, not a soak test, not auth-bypass research beyond the four-header matrix already documented.

In scope

- GET (and code-level HEAD allowance on the proxy) against `/v1/baseball/*`.
- Happy path on sample IDs from search → team events → lineups → seasons.
- Status filters, pagination edges already in `probe_variations.py`, validation 400/404.
- Product workflows on the pages in `App.tsx`.

Out of scope

- Load/stress, fuzzing, destructive writes (proxy returns **405** for non-GET/HEAD — **code**, not executed here).
- Sports other than baseball (middleware 400 if path is not `/v1/baseball`).
- Certifying production SLOs or completeness of global baseball coverage.

How tests were executed (existing)

1. `scripts/probe_all_endpoints.py` discovers IDs via search (Yankees / Ohtani / MLB), team `events/last`, lineups, seasons, enrichments; calls each of 42 templates; writes `data/baseball-endpoint-evidence.json`.
2. Same script's `extraProbes` (11): finished/inprogress, page 9999, page 2, search missing q, event abc / 999999999, standings home/away, player/team `events/next`.
3. `scripts/probe_variations.py` — 51 tests (states, second team/player, page 2000, limits, status tokens, enrichment types).
4. Full-catalog lab → `data/full-catalog-test.json` (71 rows) and `data/calls.json` (196 calls).
5. This review: additional **read-only GETs** listed in section O (auth matrix, identity of fixtures vs events, filters, detail R/H/E, h2h/best-players/top-performers/rankings, standings, players page 9999, search without q, not-in-catalog season stats).

Environments: probes hit `MSDATA_BASE_URL` directly with both key headers. The UI hits `http://127.0.0.1:5175/msdata/...` (Vite injects the same header names). Additional GETs used curl + both headers except the auth matrix.

ID chain (documented and used): search `q=Yankees` → team `84917477` → `events/last` finished row → event `67599823`; search Ohtani → `71698966`; search MLB → `53168725`; seasons → `68144200`; lineups → manager `62643844`, player `39625815`.

G. Detailed manual test cases

Legend: **Executed** = JSON evidence and/or this review's GET. **Expected** is the product/catalog expectation, not an SLA.

G1. Catalog routes (42)

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-C-01	`GET /event/{eventId}`	Finished sample detail	event 67599823	GET	200 match with teams, status, scores	200, 4959 B; Yankees–Rays; status object finished; inningsBaseball 3-4	PASS	evidence + this GET	Nested uniqueTournament id 53168725; `tournament.id` null
TC-C-02	`GET /event/{id}/at-bats`	Play-by-play	same	GET page=1 limit=5	paginated at-bats	200, n=5, total=70, pages=14	PASS	evidence	Canceled/not-started: empty 200 (variation)
TC-C-03	`GET /event/{id}/best-players`	Optional MOTM	same	GET	usable list or empty 200	**404** ENRICHMENT_NOT_FOUND	FAIL	evidence + this GET	Catalog includes the route
TC-C-04	`GET /event/{id}/enrichments`	Type index	same	GET	types[]	200, 5 types	PASS	evidence	
TC-C-05	`GET /event/{id}/enrichments/{type}`	h2h_events	type from index	GET h2h_events	paged events	200, total=219	PASS	evidence	Also highlights/series/team_streaks/votes 200 (variation)
TC-C-06	`GET /event/{id}/h2h`	Prior meetings	finished game	GET	duels or documented empty	200, both duels **null**, 36 B	FAIL	evidence + this GET	HTTP 200 is not an error; product H2H tab is blank
TC-C-07	`GET /event/{id}/lineups`	Roster	same	GET	home/away players	200; home dressedPlayers 16; manager null	PASS	evidence	Baseball roster, not XI=11
TC-C-08	`GET /event/{id}/player-statistics`	Box	same	GET	paged stats	200, total=32	PASS	evidence	
TC-C-09	`GET /event/{id}/player/{playerId}/statistics`	One player match stats	lineup player 39625815	GET	200 stats	200	PASS	evidence	Not season stats

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-C-10	`GET /event/{id}/pregame-form`	Form	same	GET	W/L form	200, home L,L,W,W,W	PASS	evidence	
TC-C-11	`GET /event/{id}/statistics`	Team comparison	same	GET	groups	200, period ALL, Fielding+Batting	PASS	evidence	
TC-C-12	`GET /event/{id}/team-streaks`	Streaks	same	GET	streak buckets	200, all arrays empty, 111 B	NEEDS INVESTIGATION	evidence	Probe PASS_PARTIAL; not-started 404 in calls.json
TC-C-13	`GET /event/{id}/top-performers`	Batters/pitchers	same	GET	list	200, 3 batters	PASS	evidence + this GET	Usable MOTM stand-in
TC-C-14	`GET /event/{id}/umpires`	Crew	same	GET	umpires	200, n=6	PASS	evidence	
TC-C-15	`GET /event/{id}/weather`	Conditions	same	GET	conditions	200, n=2	PASS	evidence	
TC-C-16	`GET /events`	Default feed	none	GET page=1 limit=5	current matches	200, total=**18**, all notstarted in samples	PASS	evidence	Business completeness vs team history is TC-X-01
TC-C-17	`GET /fixtures/today`	Today	none	GET	today's games	200, total=18; **same IDs as /events** (limit=20 this review)	NEEDS INVESTIGATION	evidence + this GET	Identical feed in this run
TC-C-18	`GET /manager/{managerId}`	Kevin Cash	62643844	GET	profile	200 name Kevin Cash	PASS	evidence	
TC-C-19	`GET /player/{playerId}`	Ohtani	71698966	GET	profile	200, team null	PASS	evidence	Thin preferred fields
TC-C-20	`GET /player/{id}/events/last`	Recent	same	GET	paged events	200, total=1253	PASS	evidence	

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-C-21	`GET /players`	Directory	none	GET	paged players	200, total=9758	PASS	evidence	
TC-C-22	`GET /search`	q=Yankees	q required	GET	typed hits	200, 1 team 84917477	PASS	evidence + this GET	
TC-C-23	`GET /team/{teamId}`	Yankees	84917477	GET	profile	200, country null	PASS	evidence	
TC-C-24	`GET /team/{id}/enrichments`	Index	same	GET	types	200, n=8	PASS	evidence	
TC-C-25	`GET /team/{id}/enrichments/{type}`	events_last	type from index	GET	payload	200, total=30	PASS	evidence	
TC-C-26	`GET /team/{id}/events/last`	History	same	GET	paged events	200, **total=2123**, pages=425	PASS	evidence + this GET	Contrasts with /events?status=finished
TC-C-27	`GET /team/{id}/featured-event`	Featured	same	GET	one event	200; findings: 67599823 finished	PASS	evidence + findings	Semantics unclear vs live
TC-C-28	`GET /team/{id}/media`	Media	same	GET	paged media	200, total=10	PASS	evidence	
TC-C-29	`GET /team/{id}/near-events`	Prev/next	same	GET	envelope	200	PASS	evidence	
TC-C-30	`GET /team/{id}/performance`	Form	same	GET	form+events	200, total=20	PASS	evidence	Scope undocumented (findings P3)
TC-C-31	`GET /team/{id}/players`	Roster	same	GET	players[]	200, n=48	PASS	evidence	Nested player id/name
TC-C-32	`GET /team/{id}/rankings`	Rankings	same	GET	list or empty 200	**404** ENRICHMENT_NOT_FOUND	FAIL	evidence + this GET	

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-C-33	`GET /team/{id}/standings-seasons`	Seasons with tables	same	GET	list	200, total=4	PASS	evidence	
TC-C-34	`GET /team/{id}/unique-tournaments`	Team competitions	same	GET	list	200, total=3	PASS	evidence	
TC-C-35	`GET /teams`	Directory	none	GET	paged teams	200, total=255	PASS	evidence	
TC-C-36	`GET /tournaments`	Directory	none	GET	paged competitions	200, total=84	PASS	evidence	
TC-C-37	`GET /unique-tournament/{id}`	MLB	53168725	GET	competition	200 name MLB	PASS	evidence	currentSeasonId missing on object (full-catalog)
TC-C-38	`GET /unique-tournament/{id}/events`	Competition feed	same	GET	paged events	200, total=29267, pages=2927	PASS	evidence	
TC-C-39	`GET /unique-tournament/{id}/scheduled-events/{date}`	Calendar 2026-10-21	date	GET	day's events	200, total=1	PASS	evidence	
TC-C-40	`GET /unique-tournament/{id}/season/{sid}/round/{round}`	Round 1	season 68144200	GET	events	200, total=2458	PASS	evidence	
TC-C-41	`GET /unique-tournament/{id}/season/{sid}/standings/total`	Table	same	GET	rows	200, 90 rows, type total	PASS	evidence + this GET	
TC-C-42	`GET /unique-tournament/{id}/seasons`	Season list	same	GET	seasons	200, total=24	PASS	evidence	

G2. Filters, pagination, errors, auth, extra route

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-X-01	`GET /events?status=finished`	Results filter	finished games exist on team feed	GET limit=5	non-zero when history exists	200, items=[], **total=0**	FAIL	evidence extra, variation, this GET	Yankees last total=2123 same window
TC-X-02	`GET /events?status=inprogress`	Live filter	—	GET	live games or honest empty if none live	200, **total=0**	FAIL	same	Cannot distinguish “no live games” vs broken index
TC-X-03	`GET /events?status=notstarted`	Upcoming filter	—	GET	upcoming	200, total=18	PASS	variation + this GET	
TC-X-04	`GET /events?status=canceled`	Canceled filter	canceled id 82011329 exists	GET	canceled games	200, total=0	FAIL	variation + this GET	
TC-X-05	`GET /events?status=postponed`	Postponed	—	GET	postponed or empty	200, total=0	NEEDS INVESTIGATION	variation + this GET	No independent postponed id in repo
TC-X-06	`GET /events?status=ended`	Invalid token	—	GET	400	400; allowed: notstarted, inprogress, halftime, pause, finished, postponed, canceled, awarded, interrupted, coverage_lost	PASS	variation + this GET	`ended` is not in the enum; detail `description` is “Ended”
TC-X-07	`GET /events` page 1 vs 2	Paging	total=18	GET page=1 then 2, limit=5	different slices	page1 first 28606880, page2 first 56340050	PASS	evidence extra	
TC-X-08	`GET /events?page=9999`	Beyond totalPages	totalPages=4	GET	honest empty	200, items=[], **total still 18**, hasMore=false	FAIL	evidence extra	Empty list is honest; leftover total is not

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-X-09	`GET /players?page=9999`	Beyond 1952	—	GET	honest empty	200, n=0, **total=9758**, totalPages=1952	FAIL	this GET (same as catalog extra)	
TC-X-10	`GET /players?page=2000`	page 2000 vs totalPages 1952	—	GET	empty 200	200, count=0	PASS	variation	No fake rows
TC-X-11	`GET /unique-tournament/53168725/events?page=2000`	page 2000 in range	totalPages≈2927	GET limit=10	real rows	200, count=10, total=29267	PASS	variation	Page 2000 is valid here
TC-X-12	`GET /events?limit=2000`	Oversize limit	—	GET	validation error	**400**	PASS	variation	Same for search limit=2000
TC-X-13	`GET /events?limit=1` / `limit=100`	Limit honoured	—	GET	n≤limit	limit=1 → n=1; limit=100 → n=18	PASS	variation	
TC-X-14	`GET /search` missing q	Validation	omit q	GET	400	400, message **array** of 4 strings	PASS	evidence + this GET	
TC-X-15	`GET /event/abc`	Non-numeric id	—	GET	400	400 string message	PASS	evidence	
TC-X-16	`GET /event/999999999`	Unknown id	—	GET	404	404 EVENT_NOT_FOUND	PASS	evidence	Contrast with enrichment 404
TC-X-17	Standings home/away	Split tables	MLB 2026 season	GET home, away	rows or documented empty	200, 1 table each, **rows: []** (59 B); total 90 rows	FAIL	evidence extra + this GET	
TC-X-18	Player/team `events/next`	Upcoming	Ohtani, Yankees	GET	upcoming or empty	both 200, total=0	PASS	evidence extra	Empty may be off-season; not a transport error

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-X-19	Auth: both headers	`/events?limit=1`	keys in env	GET	200	200, keys items/page/limit/total	PASS	this GET	Values not recorded
TC-X-20	Auth: `X-Live-Key` only	same	—	GET	documented contract unknown	**200**	PASS	this GET	Confirms improvements.md
TC-X-21	Auth: `x-apim-key` only	same	—	GET	unknown	**403** Forbidden JSON	FAIL	this GET	Fail vs “either key works”; contract missing
TC-X-22	Auth: no key	same	—	GET	401/403	**403**, body not JSON (5490 B)	PASS	this GET	Non-JSON error vs JSON 403 with APIM only
TC-X-23	Season stats **not-in-catalog**	Ohtani MLB 2026	player/tour/season ids	GET football-style path	unknown / catalog says absent	**200**, battingAvg 0.275, HR 30, AB 523, appearances 135	PASS	this GET	Not wired in `Player.tsx`
TC-X-24	Invalid direction `events/sideways`	Enum	Ohtani	GET	400	400 “enum string is expected”	PASS	full-catalog / calls	
TC-X-25	Invalid standings `form`	Enum	MLB season	GET	400	400	PASS	full-catalog	
TC-X-26	Invalid calendar date	`not-a-date`	MLB	GET	400	400 expected YYYY-MM-DD	PASS	full-catalog	
TC-X-27	Event states: canceled / notstarted	Same sub-routes	ids 82011329, 28606880	GET detail, lineups, statistics, at-bats, top-performers, best-players	documented empty vs 404	detail/lineups/statistics/top-performers 200; at-bats empty 200; best-players 404 all three states	PASS	variation (18 tests)	Empty vs unsupported still undocumented
TC-X-28	Second entities	Rays / Volpe last	team 20869353, player 39625815	GET events/last	200	Rays total=2081; Volpe total=615	PASS	variation	

G3. Product workflows (UI)

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-P-01	Home Live	Open `/`	`pnpm dev` :5175	Load Home	Live cards if games in progress	Code filters events to Live; API inprogress total=0 → empty “No matches.” Lab Home also N+1’d event detail at 12:58Z	FAIL	Home.tsx; this GET; calls.json Home	Current source has no N+1
TC-P-02	Home Today cards	Open `/`	same	View Today	R/H/E + competition name	List series scores; competitionName string; findings HOME_MATCH_SCORE_MISSING	FAIL	evidence list row; findings	MatchCard prints series via `scoreDisplay`
TC-P-03	Matches Results fallback	Open `/matches?tab=results`	finished filter empty	Click Results	Completed games	`calls.json` : `/events?status=finished` 200 73 B then `/team/84917477/events/last` fallbackUsed	FAIL	Matches.tsx; calls; findings EVENT_FINISHED_FILTER_INCOMPLETE	Yankees-only
TC-P-04	Match centre sample	`/match/67599823`	finished id	Open tabs	Header + extras	Detail 200; best-players tab will show HTTP 404 (JsonPanel); h2h null JSON	FAIL	Match.tsx; this GET	FAIL is for best-players/h2h tabs; overview PASS (see C-01)
TC-P-05	Search Yankees	`/search`	—	Submit Yankees	Team link	200, id 84917477, type team (`calls.json` Search)	PASS	calls + this GET	
TC-P-06	Search Ohtani / MLB	`/search`	—	Submit	Player / competition	Catalog audit 200 (249 B / 444 B). UI Search screen only logged Yankees	PASS	calls.json Full catalog audit	UI grouping of those queries `NOT TESTED` on Search page
TC-P-07	Search typo `Yankeess`	playbook	—	Submit	empty or near-miss	No JSON / calls row	NOT TESTED	—	Playbook-only
TC-P-08	Player Ohtani	`/player/71698966`	—	Open	Profile + recent	Overview 200 (`calls.json` Player). Season stats not requested by page	PASS	Player.tsx; calls	Season line absent by catalog, not by API (TC-X-23)
TC-P-09	Team Yankees rankings tab	`/team/84917477?tab=rankings`	—	Open tab	Rankings or empty	GET 404	FAIL	Team.tsx; this GET	

ID	Endpoint / workflow	Scenario	Preconditions	Steps	Expected	Actual	Status	Evidence	Notes
TC-P-10	Competition standings home/away	`/competition/53168725?tab=standings`	season selected	Toggle home/away	Tables	API empty rows; UI JsonPanel	FAIL	Competition.tsx; this GET	total PASS (TC-C-41)
TC-P-11	Manager from lineup	`/manager/62643844`	id from lineup	Open	Profile	API 200. No calls.json Manager screen	NOT TESTED	ManagerPage source; API C-18	Header has no Manager link
TC-P-12	Endpoint lab	`/endpoints`	sample IDs in form	Click each path	42 GETs	Source maps CATALOG; IDs match SAMPLE + 62643844 / 68144200	NOT TESTED	EndpointLab.tsx	Covered at API layer by probe, not as this page in calls.json
TC-P-13	Proxy methods	`/msdata`	local/Vercel	Non-GET	405	**Code** `middleware.ts` / API handler	NOT TESTED	source only	Not executed (non-destructive policy)

H. Pagination analysis

Observed pagination fields: `page`, `limit`, `total`, `totalPages`, `hasMore`. Not every list uses them (roster `players[]`, standings, umpires, top-performers).

Observation	Evidence	Result
Page 2 is a new slice on `/events`	extra probe: first ids 28606880 vs 56340050	PASS
Directories page 2 differ	full-catalog teams/players/tournaments `different=true`	PASS
`page > totalPages` → HTTP **200**, empty array, **`total` unchanged**, `hasMore: false`	events page 9999 (total 18 / pages 4); players page 9999 and 2000 (total 9758 / pages 1952); team events page 2000 (total 2123 / pages 425); at-bats page 2000 (total 70 / pages 14)	FAIL as a client-stop contract; PASS that rows are not fabricated
Page 2000 **inside** range returns real rows	MLB unique-tournament events page 2000: n=10, totalPages 2927, total 29267; last page 2927 n=7; page 2928 empty	PASS
`limit=2000` rejected	events and search → 400 (variation EXPECTED)	PASS
`limit=1` and `limit=100` honoured	n=1; n=18 (=total)	PASS
Search huge page	full-catalog page 9999, total stays 1, items empty	same leftover-total pattern

Observation	Evidence	Result
UI paging	Players/Teams use `hasMore` only (`Player.tsx` , `Team.tsx`)	Safe against leftover `total` , but only if `hasMore` is false (it was on page 9999)

I. Error handling and HTTP status

HTTP	When observed	Body shape	Product handling
200	Most catalog GETs	Resource JSON	`msdata.ts` parses JSON; `error` only if `!response.ok`
200 empty	filters total=0; h2h nulls; events/next; standings home/away rows=[]	Valid JSON	Matches Results treats empty finished as fallback; Home Live shows “No matches.” H2H tab still 200 so JsonPanel shows nulls, not DataState error
400	search missing/short `q` ; event/abc; events status=ended; events/sideways; standings/form; not-a-date; limit=2000	`error: "Bad Request"` ; `message` string **or** array	Search shows concatenated/stringified message (`calls.json` error field joins array with commas)
403	`x-apim-key` only; no key	APIM-only: JSON Forbidden; no-key: **non-JSON** 5490 B	UI copy in `msdata.ts` for 401/403 mentions setting `MSDATA_API_KEY` and `MSDATA_LIVE_KEY` in Vercel — does not say Live-Key-only is sufficient
404 entity	unknown event/player/team numeric id	`details.code` EVENT_NOT_FOUND / equivalent	Distinguishable message (“Baseball match … was not found”)
404 enrichment	best-players, rankings; team-streaks on not-started 28606880	`details.code` ENRICHMENT_NOT_FOUND, `reason` no_data_recorded	Same HTTP as missing entity; Match/Team tabs look broken
404	(none in catalog sweep for missing `/msdata` proxy)	—	`msdata.ts` maps 404 to “ `/msdata` proxy not found” **even for API enrichment 404s** — product copy can mis-blame the proxy
405	non-GET/HEAD on proxy	text “Method not allowed”	Code only; not executed
5xx	—	—	**Not observed** in calls.json or probes

`widgetCopy` in `DataState.tsx` knows `OPTIONAL_404` , but Match/Team tabs always render `JsonPanel` for extras rather than mapping 404 vs empty.

J. Product-level manual validation

Method: source review of every `App.tsx` route, plus `data/calls.json` screens (Home 77, Full catalog 93, Match Centre 15, Team 4, Competition 3, Matches 2, Search 1, Player 1). `docs/baseball_api_improvements.md` (checked 21 Oct 2026) is treated as a **documented UI walk**, not as new screenshots.

Journey	Can a user complete it?	Evidence
Browse today's slate	Partially: 18 upcoming cards, series scores, "MLB Postseason" via `competitionName`	Home/Matches Today
See live games	No — Live empty while filter returns total=0	TC-P-01, TC-X-02
See completed games globally	No — Results empty then Yankees-only fallback	TC-P-03
Open a finished game and read R/H/E	Yes for `67599823`	TC-C-01
At-bats / lineups / box / umpires / weather / top-performers	Yes on finished sample	catalog 200s
Best players tab	No — 404	TC-C-03
H2H tab	No useful data — null duels	TC-C-06
Search Yankees and open team	Yes	TC-P-05
Open Ohtani and see season AVG/HR	No in UI (route exists off-catalog)	TC-P-08, TC-X-23
Yankees roster	Yes (48 players)	TC-C-31
Yankees rankings	No — 404	TC-C-32
MLB total table	Yes (90 rows)	TC-C-41
MLB home/away tables	No — empty rows	TC-X-17
Manager profile	API yes; chrome does not link it	TC-P-11
Click all 42 routes	Page exists; API already swept	TC-P-12

Consistency across pages: `asMatch` / `MatchCard` are shared. List feeds without a tournament object show "Competition unknown" when `tournament.id` and `name` are missing; fixtures rows have `competitionName` but `asMatch` reads `uniqueTournament` / `tournament` / `tournamentName`, **not** `competitionName` — Home cards can show "Competition unknown" even when the API sent `competitionName`. That is a **client mapping gap** observed in source, not a separate API test.

Edge states checked in evidence: empty Live; empty Results then fallback; canceled first row on Yankees last; not-started at-bats empty; unknown ids 404; missing search q 400. **Not tested:** authenticated vs anonymous UI (keys are server-side); mobile viewport; `/endpoints` click-through as a user; Search typo; Manager navigation from a lineup click (lineup JsonPanel is raw JSON, not links).

K. Findings / defect register

Severity is product impact in this lab, not a vendor SLA.

ID	Severity	Area	Finding	Evidence	Status
F-01	**Critical**	`/events` filters	`status=finished` and `inprogress` return total=0 while team `events/last` has 2123 games. Live and Results cannot be built from the global feed.	extra probes, variation, this GET, Matches.tsx fallback, findings EVENT_FINISHED_FILTER_INCOMPLETE	Open

ID	Severity	Area	Finding	Evidence	Status
F-02	**High**	List vs detail schema	List: string status + `{ series }` + `competitionName`. Detail: status object + `inningsBaseball` R/H/E + nested uniqueTournament. Home cannot print a baseball scoreboard from the list.	evidence previews; this GET; findings LIST_SCORE_SERIES_ONLY / SCHEMA_INCONSISTENCY / HOME_MATCH_SCORE_MISSING	Open
F-03	**High**	Optional enrichments	`best-players` and team `rankings` 404 on valid finished Yankees–Rays / Yankees. `top-performers` is 200. Same 404 class as unknown entity.	evidence + this GET	Open
F-04	**High**	Auth contract	`X-Live-Key` alone 200; `x-apim-key` alone 403; both 200; none 403 non-JSON. No written rule in `.env.example` beyond setting both names.	this GET; improvements.md	Open
F-05	**High**	Catalog gap	Player season statistics path returns 200 with battingAvg/HR/AB and is not in the 42 routes or `Player.tsx`.	this GET; catalog.ts	Open
F-06	**Medium**	`/h2h`	200 with null duels on finished sample.	evidence + this GET	Open
F-07	**Medium**	Standings home/away	200 empty `rows` vs total 90.	extra probes + this GET	Open
F-08	**Medium**	Pagination	`page > totalPages` is 200 empty with **stale high `total`**.	extra probes, variation, this GET	Open
F-09	**Medium**	`/fixtures/today` vs `/events`	Identical 18 IDs and byte size in this run. Unclear whether “today” is a calendar slice.	this GET	Open
F-10	**Medium**	Tournament id on detail	`tournament.id` null; nested `uniqueTournament.id` 53168725. `asMatch` prefers `tournament` so competition links break.	this GET; msdata.ts; findings BROKEN_RELATIONSHIP	Open (API + client)
F-11	**Medium**	`/events?status=canceled`	total=0 while canceled event 82011329 is first on Yankees last.	this GET; variation	Open
F-12	**Medium**	Client 404 copy	`msdata.ts` treats any HTTP 404 as “`/msdata` proxy not found”. Enrichment 404s get the wrong operator hint.	msdata.ts vs 404 bodies	Open (client)
F-13	**Low**	Error `message` type	Array vs string vs `{code,message}` (documented in improvements; search 400 array confirmed).	this GET search; ended 400 array	Open
F-14	**Low**	Player/team preferred fields	Ohtani `team` null; Yankees `country` null on team resource; findings P3 list.	evidence previews; findings	Open
F-15	**Low**	Featured-event semantics	Returns finished 67599823; findings FEATURED_EVENT_SEMANTICS_UNCLEAR.	findings; probe 200	Open

ID	Severity	Area	Finding	Evidence	Status
F-16	**Low**	Lineup manager	Home `manager` null; away manager used for `/manager/62643844`.	evidence lineup preview; improvements.md	Open
F-17	**Low**	`competitionName` unused	List rows include `competitionName`; `asMatch` does not read it → "Competition unknown".	msdata.ts vs list keys	Open (client)

`data/findings.json` also records `SLOW_ENDPOINT` (e.g. `/events` 1174ms, some event details >800ms). Those are lab timings, not load tests; not promoted to this register as performance defects.

L. Usability / integration assessment

No numeric scores.

What integrates cleanly: search hits with `type` + `id` + `name`; event detail R/H/E; at-bats; lineups with batting order; player-statistics; umpires; weather; top-performers; team roster; media; near-events; MLB total standings; seasons list; manager by id; pagination `hasMore` on directories.

What forces workarounds: Results tab hard-codes Yankees `SAMPLE.teamId`; Home Live depends on a global inprogress filter that is empty; MatchCard must interpret string-or-object status and series-or-innings scores (`statusOf`, `scoreDisplay` in `msdata.ts`).

What looks like a broken page: best-players and rankings tabs (404 JSON); h2h tab (nulls); standings home/away (empty table JSON); Live (empty state).

Lab vs current source: on 21 Oct 12:58–13:01Z the Home lab recorded N+1 `GET /event/{id}` (`fallbackUsed: true`). Current `Home.tsx` only calls three list endpoints. Do not assume the N+1 still ships; do assume list scores are still series-only.

Operator experience: `.env.example` asks for both key names. Vite sends both if present. Live-Key alone is enough on `/events?limit=1`; APIM alone is not. Customers copying only the APIM key will see 403.

Accessibility / visual QA: not in scope. Match extras are raw `JsonPanel` (`JSON.stringify`), not baseball-specific widgets — fine for a lab, not a consumer UI.

M. Coverage summary (how counts were calculated)

Bucket	Count	How it was counted
Catalog routes	**42**	`len(CATALOG)` in `catalog.ts`; `data/baseball-routes.json` `"count": 42`; same list in both files
Catalog routes executed in 12:31Z probe	**42 / 42**	`baseball-endpoint-evidence.json` `"routeCount": 42` and `len(routes)==42`
Probe extra probes	**11**	`len(extraProbes)` in that JSON

Bucket	Count	How it was counted
Variation tests	**51**	`summary.totalTests` in `baseball-variation-tests.json` (PASS 31, PASS_EMPTY 14, EXPECTED 6)
Full-catalog lab rows	**71**	`rowCount` / `len(rows)` in `full-catalog-test.json` ; `catalogCovered` 42, `uncovered` []
Recorded HTTP calls	**196**	`len(calls.json)` ; statuses 181×200, 8×400, 7×404
UI screens with ≥1 calls.json row	7 named screens + catalog sweep	Counter of `screen` ; **Manager** and **Endpoint lab page** have no screen rows
Additional GETs this review	Auth 4 header combos + 21 path GETs listed in O	curl, GET only
Catalog cases in §G1 marked executed	42	All have probe evidence
§G1 PASS / FAIL / NEEDS INVESTIGATION	37 PASS, 3 FAIL (C-03, C-06, C-32), 2 NEEDS INVESTIGATION (C-12, C-17)	From §G1 Status column
Product journeys in §J	14 rows	One per journey; mix of PASS/FAIL/NOT TESTED
Load/stress	**0**	Explicitly not run
Destructive requests	**0**	GET/HEAD only

This is not 100% product certification. Gaps: Manager UI walk, `/endpoints` click-through, Search typo, Search Ohtani/MLB on the Search **page** (only catalog audit), Team tabs beyond overview/featured/near/performance in `calls.json` , Match tabs not all present in Match Centre lab rows, incidents/injuries uncalled, POST 405 unexecuted, postponed filter without a known postponed id, whether inprogress=0 means no live baseball worldwide.

N. Recommendations and conclusion

Must

1. Make `/events?status=finished` and `inprogress` return the games that already exist on team/player `events/last` (F-01). Until then, Live/Results are not shippable without a team-specific hack.
2. Put detail-equivalent **status object** and **`inningsBaseball` (R/H/E)** on `/events` and `/fixtures/today` rows (F-02).
3. Return **200 + empty** for missing optional enrichments (`best-players` , team `rankings`) instead of 404, **or** drop them from the baseball catalog and point MOTM at `/top-performers` (F-03).
4. Publish one auth rule: which of `X-Live-Key` / `x-apim-key` is required (F-04).

Should

5. Add `GET /v1/baseball/player/{id}/unique-tournament/{tid}/season/{sid}/statistics` to the baseball catalog and to the player page (F-05) — it already returns battingAvg / HR / AB.
6. Populate `/h2h` or document null duels as the empty contract (F-06).
7. Fill standings `home` / `away` or remove those types from the product tabs (F-07).

8. For `page > totalPages`, either 400 or keep 200 but do not leave `total` looking like a full feed (F-08).
9. Document `/fixtures/today` vs `/events` (F-09).
10. Set `tournament.id` (or tell clients to use nested `uniqueTournament.id`); optionally fix `asMatch` to prefer nested id when `tournament.id` is null (F-10).
11. Stop mapping API 404 bodies to "proxy not found" in `msdata.ts` (F-12).

Nice-to-have

12. One `message` type on errors (F-13).
13. Player.team and team.country on profile resources (F-14).
14. Document featured-event vs near-events (F-15).
15. Manager id on both lineup sides; document roster size (F-16).
16. Map list `competitionName` in `asMatch` (F-17).
17. `hasMore`-only paging is already a reasonable client defense.

Conclusion

The Baseball Product is a thin, honest lab over **42** deployed GETs. Enough of the catalog works to demonstrate search, a finished match centre, a team roster, and an MLB table. **Live and Results do not**, because the global `/events` status index does not match team history. List scoreboards are series-only; optional extras 404; season stats exist off-catalog. Treat this document as a **21 Oct 2026 evidence pack**, not a go-live certificate.

O. Appendix / evidence paths

O1. Repo paths

Path	Use
`apps/web/src/catalog.ts`	42 route strings
`apps/web/src/msdata.ts`	Client, SAMPLE ids, status/score mapping
`apps/web/src/App.tsx`	Routes
`apps/web/src/pages/*.tsx`	Screen → endpoint wiring
`apps/web/vite.config.ts`	`/msdata` proxy, header names
`apps/web/api/msdata/[...path].ts`, `apps/web/middleware.ts`	Production proxy (GET/HEAD only)
`data/baseball-routes.json`	Official 42
`data/baseball-endpoint-evidence.json`	Full probe (large; do not paste)
`data/baseball-variation-tests.json`	51 variation tests
`data/full-catalog-test.json`	71 lab rows
`data/calls.json`	196 call log
`data/findings.json`, `data/gaps.json`	Structured findings
`docs/baseball_api_testing_report.md`	Generated 42-route report (kept)

Path	Use
`docs/baseball_api_variation_testing.md`	Variation report (kept)
`docs/baseball_api_endpoint_catalog.md`	Shape catalog (kept)
`docs/baseball_api_improvements.md`	API-team list (kept)
`docs/Baseball_Product_Review.md`	Playbook (kept)
`docs/baseball_api_reports.md`	Combined lab notes (kept; contains scores not used here)
`scripts/probe_all_endpoints.py`, `scripts/probe_variations.py`	Repeatable GET probes
`.env.example`	Config **names** only

O2. Existing evidence vs additional tests this review

Existing (preferred, used first): all files in O1 generated 21 Oct 2026 12:31–13:01Z, plus source.

Additional read-only GETs this review (after those files): GET only; no bodies committed; secrets not logged.

Call	Result (sanitized)
`/events?limit=1` headers: both / Live-Key only / APIM only / none	200 / **200** / **403** Forbidden JSON / **403** non-JSON 5490 B
`/events?page=1&limit=20` and `/fixtures/today?page=1&limit=20`	both 200, 5290 B, **same 18 IDs**, total=18
`/events?status=` finished, inprogress, notstarted, canceled, postponed, ended	200 total=0; 200 total=0; 200 total=18; 200 total=0; 200 total=0; **400** enum list
`/team/84917477/events/last?limit=5`	200, total=2123; row0 id 82011329 canceled, series only
`/event/67599823`	200, 4959 B; status finished; inningsBaseball 3–4; venue Yankee Stadium
`/event/67599823/h2h`	200, both duels null
`/event/67599823/best-players`	404 ENRICHMENT_NOT_FOUND
`/event/67599823/top-performers`	200, 3 batters
`/team/84917477/rankings`	404 ENRICHMENT_NOT_FOUND
standings total / home / away	200 × 90 rows / 200 × 0 / 200 × 0
`/players?page=9999&limit=5`	200, n=0, total=9758, totalPages=1952
`/search` without q	400, message array
`/search?q=Yankees`	200, team 84917477
`/player/71698966/unique-tournament/53168725/season/68144200/statistics`	**200 not-in-catalog**, battingAvg 0.275, HR 30, AB 523, appearances 135

Python `urllib` on this machine failed TLS (`CERTIFICATE_VERIFY_FAILED`); additional GETs used `/usr/bin/curl` instead. Probe scripts in-repo already succeeded at 12:31Z.

O3. Sample IDs (verified)

```
{
  "eventId": 67599823,
  "teamId": 84917477,
  "playerId": 71698966,
  "lineupPlayerId": 39625815,
  "managerId": 62643844,
  "tournamentId": 53168725,
  "seasonId": 68144200,
  "canceledEventId": 82011329,
  "notStartedEventId": 28606880
}
```

O4. How to regenerate probes (does not overwrite this report)

From repo root, with `.env` containing `MSDATA_BASE_URL`, `MSDATA_API_KEY`, `MSDATA_LIVE_KEY`:

```
python3 scripts/probe_all_endpoints.py
python3 scripts/probe_variations.py
```

Those commands refresh `data/*.json` and, if you also run the generate_*scripts, the other** docs — not this file.